

// PRACTITIONER GUIDE |

Research at Terminal Velocity

Claude Code & opencode in applied research: the agent carries the process, the researcher keeps the judgment.



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// WORKSHOP GUIDE

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// WHY A DIVISION OF LABOR AT ALL

19%

SLOWER WITH AI TOOLS — MEASURED, IN METR'S RANDOMIZED TRIAL OF 16
EXPERIENCED DEVELOPERS

They had predicted a 24% speedup. Afterward, they still **believed** they got 20%. Undivided labor is the failure mode.

// WHAT YOU'LL LEAVE WITH

One rule, two tools, one loop, one Monday

- **01** — the division of labor: judgment stays home, housekeeping goes to the agent
- **02** — two tools: Claude Code and opencode, configured as research assistants
- **03** — the loop, re-divided: the agent runs the stages, you sign the gates
- **04** — a Monday on-ramp with the numbers to justify it

// 01

The division of labor

// THE ONE RULE EVERYTHING ELSE FOLLOWS FROM

Judgment stays. Housekeeping goes.

Yours to keep

- The **question** and the decision at stake
- **Converging**: filtering candidate hypotheses by rubric — scored by you, never by the model grading itself
- The three **gate signatures** — a gate "the team" owns is a gate nobody owns
- **Interpretation**: you can explain the result without opening the transcript

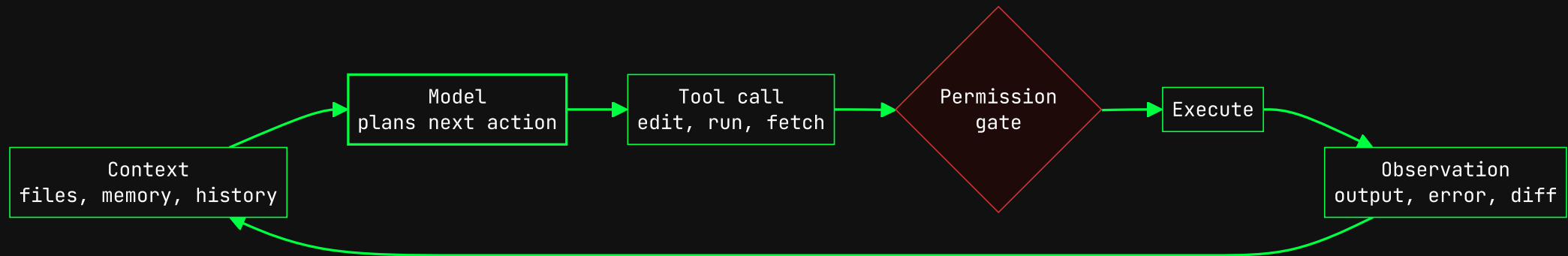
The agent's to carry

- **Diverging**: fifteen candidate hypotheses, each with mechanism and refuter
- **Executing the frozen plan**, printing every filter's row count
- **Gate paperwork**: plan restatements, checklists, critic passes
- **The logs**: transcripts, ledger hooks, a commit per stage, packaging

The book's line for it: "**The agent did the labor; the judgment stayed home.**" Less friction on you, more rigor from it.

// WHY THE HANDOFF IS SAFE

The agent loop: every action is a gated tool call



// LOG KEEPING, DELEGATED

The lab notebook writes itself. You contribute five lines.

ARTIFACT	ANSWERS	KEPT BY
Session transcript	"what did the agent do"	the tool, automatically — sessions persist and resume
Git history	"what changed, and when"	the agent — one commit per loop stage; the diff is the audit
DECISIONS.md	"what did the human choose, and why"	you — five lines per session, the one artifact only a human can write

Provenance used to be the tax on rigor. With the agent keeping two of the three ledgers, your entire logging duty is **five human-written lines per session**.

// 02

Two tools, one contract

Claude Code: the method, enforced

Built-in checkpoints you don't have to remember

- **CLAUDE.md** is read at every session start — the research contract cannot be forgotten
- **Plan mode**: the agent reads and proposes but is mechanically prevented from editing; your approval to exit is a miniature Gate B
- **Headless claude -p**: the agent as a Unix tool — literature sweeps, nightly make verify across every project

Lab infrastructure, not prompting: skills encode your SOPs; a read-only reviewer subagent runs critic passes without contaminating the builder's context; a post-edit hook appends every file change to the ledger — the notebook writes itself.

opencode: the same anatomy, open and provider-free

What the open instrument buys research

- **Provider-agnostic:** Anthropic, OpenAI, Google, or a local model server — the tool choice stops being a model choice
- **Sensitive data pattern:** identical workflow, swap the provider per project — with local models, data never leaves the building
- **Inspectable:** the whole harness is readable source — answer governance questions by reading, not trusting

Same muscles, different skin:

AGENTS.md memory, a plan-mode toggle ("suggest, don't edit"), MCP, shareable sessions for supervision, headless opencode run. Every lab in the book translates in one line.

Decision rule: Claude Code when you want the deepest workflow customization; opencode when data governance is a hard constraint — decided **before** any lab, because "discovering it after the data has left the building is not a recoverable

// WHERE THE HANDOFF HAPPENS

The contract file: housekeeping, delegated in writing

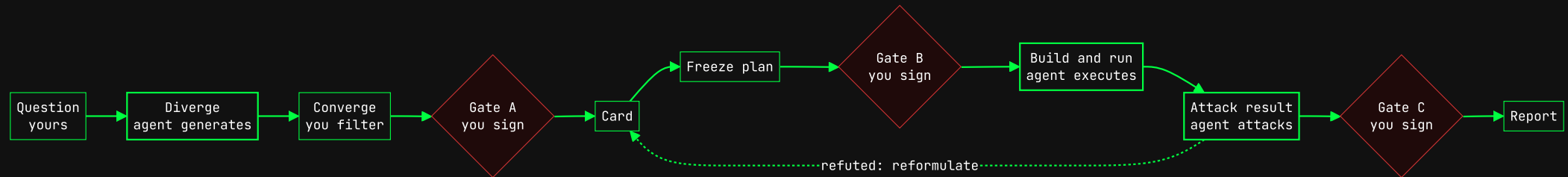
```
# CLAUDE.md -- research contract (copy to AGENTS.md)
## Hard rules
- analysis/PLAN.md is READ-ONLY: execute exactly;
  if a step is impossible, STOP and report.
- data/raw/ is read-only; outputs go to results/;
  exploratory work goes to explore/, labeled EXPLORATORY.
- Every numeric claim names the script that produced it.
- Never fabricate a citation; sources only from refs/.
## Session discipline
- One loop stage per session; commit at stage end.
- Append 5 lines to DECISIONS.md before ending.
```

// 03

The loop, re-divided

// THE AGENT RUNS THE STAGES, YOU SIGN THE GATES

Eight stages, three signatures



// YOUR JUDGMENT, FROZEN FOR THE AGENT

Pre-registration is a prompt

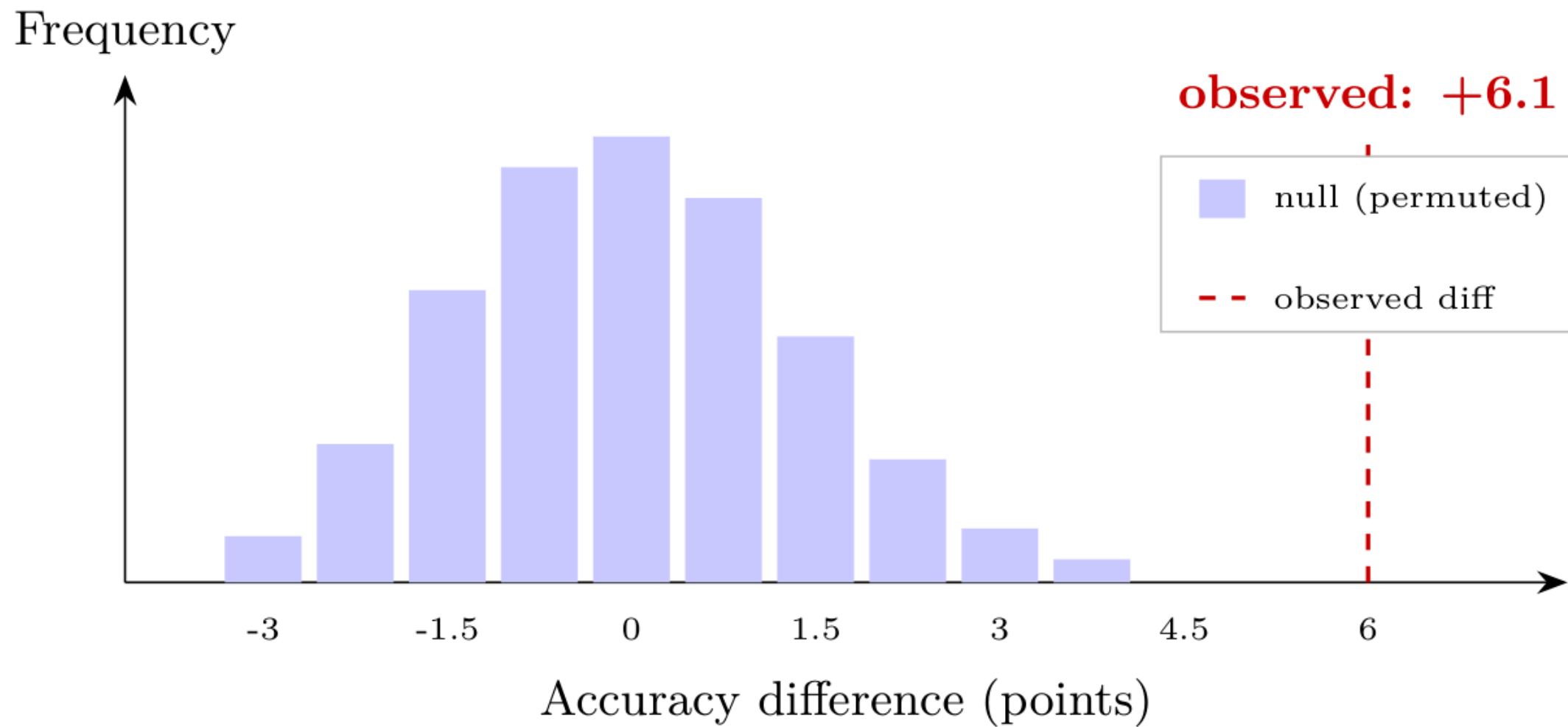
```
# Hypothesis card -- H1 (frozen 2026-07-02, commit a3f1)
H1: Retrieval-augmented triage improves top-1 routing
    accuracy by  $\geq 5$  points over the fine-tuned baseline.
Primary metric: top-1 accuracy, n = 4,000 held-out.
Test: two-sided permutation test, 10,000 resamples.
Exclusions: tickets with empty body (logged count).
Refutation: diff < 5 pts, or CI includes 0.
EXPLORATORY analyses go to explore/ -- never here.
```

// WHAT PUSHING THE PROCESS TO THE AGENT LOOKS LIKE

Day-one prompt vs contract-shaped prompt

```
# Day one
Analyze the ticket data and tell me if retrieval helps.
```

```
# Contract-shaped
Stage 6 session (build & run) for H1.
Load analysis/PLAN.md and restate it in three lines;
wait for my OK before writing code.
Then execute exactly: exclusions (log counts), top-1
accuracy for both systems, permutation test, bootstrap
CI, figures to results/fig/. Print every filter's row
count as you go. If any step is impossible, stop and
say why. Anything else you notice goes to a NOTES
section labeled EXPLORATORY.
```



// the frozen plan, executed by the agent – null distribution vs observed +6.1 (Figure 8.1, Abualhaol 2026, p.35)

// THE AGENT ATTACKS, YOU INTERPRET

The robustness pass that saved a retraction

+7.2

CLEARED THE PRE-REGISTERED TEST — THEN A STRATIFIED
CHECK FOUND THE WHOLE EFFECT IN NEAR-DUPLICATE
TICKETS: LEAKAGE, NOT LEARNING

+5.4

THE CORRECTED RUN: SMALLER, HONEST, STILL PAST THE
MINIMUM EFFECT

The agent found the leak **in four minutes** once asked. Deciding what it meant — leakage, not learning — took a researcher. That's the division working.

// PACKAGING IS HOUSEKEEPING TOO

Reproducibility: the agent's job, under an hour

RERUN

Independent rerun

another human, another machine: the only layer that tests instructions, not memory

RUN

One-command run

make reproduce && make verify, from a clean clone

DATA

Immutable data + recorded seeds

read-only raw/, every seed from one config

CODE

Versioned code + pinned environment

git + lockfile / container

Over half of surveyed researchers reported failing to reproduce **their own** experiments (Nature survey, n=1,576). Not a moral failing: reproduction was expensive friction. The agent absorbs it — packaging drops from a day to **under an hour**.

// 04

Start Monday

// WHAT THE DIVISION BUYS

The friction the agent absorbs, measured

<1 hr

REPRODUCIBILITY PACKAGING THAT
USED TO COST A DAY

5 lines

YOUR ENTIRE LOGGING DUTY PER
SESSION — THE AGENT KEEPS THE
REST OF THE LEDGER

1 pm

ONE AFTERNOON VERIFIES THE 10-15
CLAIMS YOUR REPORT LEANS ON

Total judgment cost per project: one card, three signatures, five lines a session. Skip the division and the measured default is METR's: **slower while feeling faster**, with quality losses exactly where the tool feels strongest.

// THE ON-RAMP, ASCENDING COMMITMENT

Monday to quarter

- **Mon** **Write the contract file**
One hard rule in CLAUDE.md / AGENTS.md. The handoff starts here.
- **Week** **Delegate one housekeeping job**
Verification ritual on the three references your draft leans on hardest.
- **Month** **One hypothesis, full loop**
Card, frozen plan, agent-run attack stage, clean-clone reproduce.
- **Quarter** **Calibrate yourself**
A week with the agent, a week without. Believe the clock.

// SOURCES

Where the numbers come from

- Abualhaol, "Research at Terminal Velocity", 2026 (the seeding book; embedded in this deck)
- Becker et al. (METR), "Measuring the Impact of Early-2025 AI on Experienced Developer Productivity", 2025
- Bastani et al., "Generative AI Without Guardrails Can Harm Learning", PNAS, 2025
- Baker, "1,500 scientists lift the lid on reproducibility", Nature, 2016
- Tool documentation: Claude Code (Anthropic, 2026) · opencode (Anomaly, 2026)

// REMEMBER THIS

Hand the agent the housekeeping.
Keep the judgment.

Less friction, more output — and believe the clock over the feeling.

Q & A

// THANKS FOR LISTENING

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